Quantum error correcting Code.

Alice   Bob.

$$\text{Message} \xrightarrow{\text{encode}} \text{Data} \xrightarrow[\text{Channel}]{\text{Transmit}} \text{Data} \xrightarrow{\text{decode}} \text{Message}$$

$$\text{Noise} \to \text{error} \to \widetilde{\text{Data}} \xrightarrow{\text{recovery}} \text{Data}$$

$0 \in (\mathbb{F}_2)$

$0000 = (0,0,0,0) \in \mathbb{F}_2^4$ error 0010

I. Classical Error Correction.

1. Encode : $(\mathbb{F}_2)^k \xrightarrow{C_c} (\mathbb{F}_2)^n$ $k < n$.

$$w \longmapsto Gw. \qquad [C_c] = G.$$

Def (distance & weight). $u, v \in (\mathbb{F}_2)^n$.

- $d_H(u,v) = \sum_{i=1}^{n} (u_i \oplus v_i)$ $\leftarrow$ $u = \begin{pmatrix} u_1 \\ u_2 \\ \vdots \\ u_n \end{pmatrix}$ $v = \begin{pmatrix} v_1 \\ v_2 \\ \vdots \\ v_n \end{pmatrix}$

- $\omega_H(u) = d_H(u, 0) = \sum_{i=1}^{n} u_i$

Ex $d_H(001, 101) = 1 + 0 + 0 = 1$

$\omega_H(101) = 2$.

- $d_H(C_c) = \min\{ w_H(u) : u \in \operatorname{im} C_c,\ u \neq 0 \}$

2. $w \overset{}{\underset{C_c}{\longmapsto}} \underset{\substack{\| \\ c}}{Gw} \Longrightarrow \tilde{c} = c + \underbrace{\varepsilon}_{\text{error}}$

3. Detect error & Recover.

$(\mathbb{F}_2)^k \xhookrightarrow[G]{C_c} (\mathbb{F}_2)^n \quad \operatorname{rk}(C_c) = k$.

Def (Parity Check Matrix).
A matrix $H \in M_{(n-k)\times n}(\mathbb{F}_2)$ s.t. $\ker H = \operatorname{im}(C_c) = \operatorname{im} G$
is called a parity check matrix of the code.
(of the encoding situation)

• H exists but NOT unique.

$$\begin{tikzcd}
0 \arrow[r] & (\mathbb{F}_2)^k \arrow[r, hook, "C_c", "G"'] & (\mathbb{F}_2)^n \arrow[r, two heads, "H"'] \arrow[drr] & (\mathbb{F}_2)^n / \operatorname{im}(C_c) \cong (\mathbb{F}_2)^{n-k} \arrow[r] \arrow[dr] & 0 \\
& & & & (\mathbb{F}_2)^{n-k}
\end{tikzcd}$$

Def (Syndrome Mapping).
$(\mathbb{F}_2)^n \xrightarrow[\mathrm{syn}_c]{} (\mathbb{F}_2)^{n-k}$ is called a syndrome map.

The map

$$
\begin{tikzcd}
(\mathbb{F}_2)^k \arrow[r, hook] & (\mathbb{F}_2)^n \arrow[r, "syn_c"] \arrow[dr, "\pi"'] & (\mathbb{F}_2)^{n-k} & \varepsilon \arrow[l] \\
& & (\mathbb{F}_2)^n / Im(C_c) \arrow[u, "\cong"'] &
\end{tikzcd}
$$

$w \longmapsto Gw$

$c = Gw$

$\tilde{c} = c + \varepsilon$

syn_c

$$syn(\tilde{c}) = syn(c + \varepsilon) = syn(\varepsilon).$$

Recover: $\tilde{c} + \varepsilon = c \oplus \varepsilon \oplus \varepsilon$

$$= c$$

$\boxed{\text{Thm}}$ (For "small" errors, we can recover correctly)

$$\omega_H(\varepsilon) \leq \left\lfloor \frac{d_H(C_c) - 1}{2} \right\rfloor$$

$\lfloor 2.314 \rfloor = 2$

$\lfloor 1.xxx \rfloor = 1.$

C_c: repetition

e.g. 0 0 0 0 0 0 0

Ⅵ. QECC

Message $\xrightarrow{\text{encode}}$ Data $\xrightarrow{\text{Trans.}}$ Data $\longrightarrow$ Message (Decode)

error $\longrightarrow$ Data $\longrightarrow$ Data (recover)

1. Encoding & decoding

Def ① An encoding is an injective & norm-preserving operation

- $C_q : \mathcal{H}^{\otimes k} \longrightarrow \mathcal{H}^{\otimes n}$ $(k<n)$

  $\mathcal{H}^{\otimes k} = \mathcal{H}^k = (\mathbb{C}^2)^{\otimes k}$, $\quad \mathcal{H} = \mathbb{C}^2$.

- $\mathrm{Im}(C_q) = \mathcal{H}^{C_q}$.

$D_q(\lambda\rho + (1-\lambda)\sigma) = \lambda D_q(\rho) + (1-\lambda) D_q(\sigma)$.

② Decoding is a convex-linear map

$$D_q : D(\mathcal{H}^{(q)}) \longrightarrow D(\mathcal{H}^k)$$

satisfying $D_q(C_q|\Psi\rangle\langle\Psi|C_q^*) = |\Psi\rangle\langle\Psi|$ $\quad \forall |\Psi\rangle \in \mathcal{H}^{C_q}$.

$\rho : \mathcal{H}^{C_q} \longrightarrow \mathcal{H}^{C_q}$

① $\rho = \rho^*$

② $\rho \geq 0$

③ $\mathrm{tr}(\rho) = 1$.

$|\psi\rangle \in \mathcal{H}$

$\downarrow$

$|\psi\rangle\langle\psi|$ $\quad D(\mathcal{H})$

• $|x\rangle \in \mathcal{H}^k$. e.g. $k=3$. $|x\rangle = |001\rangle$
$= |0\rangle \otimes |0\rangle \otimes |1\rangle$
$= \begin{pmatrix}1\\0\end{pmatrix} \otimes \begin{pmatrix}1\\0\end{pmatrix} \otimes \begin{pmatrix}0\\1\end{pmatrix}$.

$$\mathcal{H}^k \xrightarrow{C_q} \mathcal{H}^n$$
$$|x\rangle \longmapsto \underline{|\overline{\mp}_x\rangle} = \underline{C_q|x\rangle}.$$

2. Error Operation

Def (Quantum Operation) on $\mathcal{H}$ is

① $K : \underbrace{D(\mathcal{H})}\longrightarrow \underbrace{D_{\leq}(\mathcal{H})}$ $\underbrace{tr(\rho)} \leq 1$.

convex-linear map.

② can be represented by $K(\rho) = \sum_{l=1}^{m} K_l \rho K_l^*$

where $K_l \in \underbrace{\mathcal{L}(\mathcal{H})}$.

③ $\sum K_l^* K_l \leq I$. but "$=$" $\Longleftrightarrow tr(K(\rho)) = tr(\rho) = 1$.

Def (Error operation)

An error operation $\mathcal{E} : D(\mathcal{H}^n) \to D_{\leq}(\mathcal{H}^n)$.

$$\mathcal{E}(\rho) = \sum_{i=1}^{n} \mathcal{E}_i \rho \mathcal{E}_i^* \quad \underbrace{\mathcal{E}_i} \in \mathcal{L}(\mathcal{H}^n).$$

error operator

Or. $\mathcal{E}_i = \sqrt{p_i}\,\widehat{\mathcal{E}_i}$.

$$\Rightarrow \mathcal{E}(\rho) = \sum_i \underbrace{p_i}\, \underbrace{\hat{\mathcal{E}}_i} \rho \hat{\mathcal{E}}_i^*$$

Set $\widehat{\mathcal{E}}_0 = 1$

Example $\underset{\sim}{\mathcal{H}}^{\otimes 1} \xrightarrow{C_q} \underset{\sim}{\mathcal{H}}^{\otimes 3} \cong \boxed{(\mathbb{C}^2)^{\otimes 3}} \cong \mathbb{C}^8$

$\mathbb{C}^2 \otimes \mathbb{C}^2 \otimes \mathbb{C}^2$

$C_q : \mathcal{H}^1 \xrightarrow{\iota} \mathcal{H}^3 \xrightarrow{A} \mathcal{H}^3$

$|\psi\rangle \mapsto |\psi\rangle \otimes |0\rangle \otimes |0\rangle$

$A = |1\rangle\langle 1| \otimes \sigma_x \otimes \sigma_x$
$+ |0\rangle\langle 0| \otimes 1 \otimes 1$

$(\,)(\,) : \mathbb{C}^2 \to \mathbb{C}^2.$

Pauli Matrix

$\sigma_0 = 1$

$\sigma_1 = \sigma_x = \begin{pmatrix} 0 & 1 \\ 1 & 0 \end{pmatrix} \begin{pmatrix} x \\ y \end{pmatrix}$

$\sigma_2 = \sigma_y = \begin{pmatrix} 0 & -i \\ i & 0 \end{pmatrix}$

$\sigma_3 = \sigma_z = \begin{pmatrix} 1 & 0 \\ 0 & -1 \end{pmatrix}$

$C_q = A \circ \iota$

$$\begin{cases} C_q |0\rangle = |000\rangle \left(\cong \begin{pmatrix} 1 \\ 0 \\ 0 \\ \vdots \\ 0 \end{pmatrix} \right) \\ C_q |1\rangle = |111\rangle \left(\cong \begin{pmatrix} 0 \\ 0 \\ \vdots \\ 0 \\ 1 \end{pmatrix} \right) \end{cases}$$

$\mathcal{E} : D(\mathcal{H}^{\otimes 3}) \to D(\mathcal{H}^{\otimes 3})$

$\mathcal{E}_{00} = 1 \otimes 1 \otimes 1 \quad \mathcal{E}_{01} = 1 \otimes 1 \otimes \sigma_x$

$\mathcal{E}_{10} = 1 \otimes \sigma_x \otimes 1 \quad \mathcal{E}_{11} = \sigma_x \otimes 1 \otimes 1.$

$\mathcal{E}_{01} |000\rangle = |001\rangle.$

3. Recovery Operation

$\tilde{\tau} = c + \mathcal{E}$.

$\tilde{\tau} + \boxed{\mathcal{E}} = c + \mathcal{E} + \mathcal{E}$

$= c$.

$\boxed{\text{Def}}$

① $R : \underline{D(\mathcal{H}^{\otimes n})} \to D(\mathcal{H}^{\otimes n})$ quantum operation which preserves the trace.

$R(\rho) = \sum_i R_i \rho R_i^*$ $\quad R_i \in \mathcal{L}(\mathcal{H}^{\otimes n})$

recovery operator

② We say an error operation $\mathcal{E}$ is <u>correctable</u> if $\exists$ recovery operation R s.t.

$\forall \rho \in \underbrace{D(\mathcal{H}^{C_q})}_{\mathrm{im}(C_q)}$, $R\left(\dfrac{\mathcal{E}(\rho)}{\underline{\mathrm{tr}(\mathcal{E}(\rho))}} \right) = \rho$.

$\mathcal{E}(\rho) \in D_{\leq}(\mathcal{H}^n)$

Theorems on determining "correctable"

② $\iff \mathrm{tr}(\mathcal{E}(\rho)) = \text{const } k_{\mathcal{E}} \in [0,1]$ $\forall \rho$.

$\boxed{\text{Thm}}$ $\mathcal{E} = \{\mathcal{E}_i\}_{i=1}^n$ is correctable

$\iff \exists\, C \in \mathrm{Mat}_{n\times n}(\mathbb{C})$ s.t. ① $C = C^*$

② $\forall |\Psi_x\rangle$ ONB of $\mathcal{H}^{C_q}$, we have

$$\langle \mathcal{E}_i \Psi_x | \mathcal{E}_j \Psi_y \rangle = C_{ij} \delta_{xy}$$

$\mathcal{H}^{\otimes k} \xhookrightarrow[C_q]{} \mathcal{H}^{\otimes n}$

$(C)_{ij}$

$|x\rangle \longmapsto |\Psi_x\rangle$

ONB

$\boxed{\text{Thm}}$ ("small" errors are correctable).

$\mathcal{E} = \{\mathcal{E}_e\}$ $\underline{u\text{-qubit}}$ error.

$$u \leq \left\lfloor \frac{d_D(C_q) - 1}{2} \right\rfloor$$. Then $\mathcal{E}$ is correctable.

* $\mathcal{E} = \{\mathcal{E}_i\}$ is $\underline{k\text{-qubit}}$ if. $\mathcal{E}_i$s are $\boxed{k\text{-local}}$.

$\ulcorner \mathcal{E}_i \in \mathcal{L}(\mathcal{H}^{\otimes n}) \cong \mathcal{L}(\mathcal{H}) \otimes \mathcal{L}(\mathcal{H}) \otimes \cdots \otimes \mathcal{L}(\mathcal{H})$

$$= \mathcal{E}_i = \sum a \underline{A_1 \otimes A_2 \otimes \cdots \otimes A_n}$$

$$\left| \{ j \in \{1, 2, \ldots, n\} : A_j \neq 1 \} \right| \leq k \lrcorner$$

e.g. $\mathcal{E} = \boxed{\sigma_x \otimes \sigma_y \otimes 1 \otimes 1} + \sigma_y \otimes 1 \otimes 1 \otimes \sigma_x$.

2-local.

4. Detect & Correct $\Rightarrow$ Syndrome

$\boxed{\text{Def}}$ $\mathcal{H}^k \xhookrightarrow{C_q} \mathcal{H}^n$ $\mathcal{E} = \{\mathcal{E}_i\}_{i \in I} \geq 0$

also $\mathcal{E}_i = \sqrt{p_i}\,\hat{\mathcal{E}}_i$, $\hat{\mathcal{E}}_i \in \mathcal{U}(\mathcal{H}^n)$

Define $\mathcal{H}^S$ to be a Hilbert space of $\dim = |I|$ (i.e. $\mathcal{H}^S \cong \mathbb{C}^{|I|}$)

and $\{|\varphi_i\rangle\}_{i \in I}$ = ONB of $\mathcal{H}^S$.

Then the syndrome operator $S_\mathcal{E}$ is a norm-preserving operator

$S_\mathcal{E} : \mathcal{H}^n \to \mathcal{H}^n \otimes \mathcal{H}^S$ s.t.

$$\boxed{S_\mathcal{E}(\hat{\mathcal{E}}_a |\Psi\rangle) = \hat{\mathcal{E}}_a |\Psi\rangle \otimes |\varphi_a\rangle}$$

$\forall \hat{\mathcal{E}}_a$, $\forall \Psi \in \mathcal{H}^{C_q}$.

- $a = \mathrm{syn}_q(\hat{\mathcal{E}}_a)$ syndrome of $\hat{\mathcal{E}}_a$.

$S_\mathcal{E}$ (What we got but don't know) $= (\;\;//\;\;) \otimes |\varphi_3\rangle$

$\hat{\mathcal{E}}_3$

Example $\mathcal{H}^1 \xhookrightarrow{C_0} \mathcal{H}^3$ $C_{\mathcal{Y}} = A\circ 2$

$$\left\{\begin{array}{l} |0\rangle \longmapsto |000\rangle \\ |1\rangle \longmapsto |111\rangle \end{array}\right.$$

errors $\mathcal{E}_{00} = 1\otimes 1 \otimes 1$ $\boxed{\mathcal{E}_{10} = 1\otimes \sigma_x \otimes 1}$

$\mathcal{E}_{01} = 1\otimes 1\otimes \sigma_x$ $\mathcal{E}_{11} = \sigma_x \otimes 1 \otimes 1$

$$S_{\mathcal{E}} : \mathcal{H}^{\otimes 3} \longrightarrow \mathcal{H}^{\otimes 3} \otimes \mathcal{H}^{\otimes 2} \quad \left(\begin{array}{l}\cong \mathbb{C}^2\otimes\mathbb{C}^2 \\ \cong \mathbb{C}^4\end{array}\right)$$

$$\begin{aligned} \| \; & (|111\rangle\langle 111| + |000\rangle\langle 000|)\otimes |00\rangle \\ & + (|110\rangle\langle 110| + |001\rangle\langle 001|)\otimes|01\rangle \\ & + (|101\rangle\langle 101| + |010\rangle\langle 010|)\otimes|10\rangle \\ & + (|100\rangle\langle 100| + |011\rangle\langle 011|)\otimes|11\rangle . \end{aligned}$$

$$|\Psi\rangle = a|000\rangle + b|111\rangle \in \mathcal{H}^{\otimes 3}$$

$$\mathcal{E}_{10}|\Psi\rangle = a|010\rangle + b|101\rangle .$$

$$S_{\mathcal{E}}(\mathcal{E}_{10}|\Psi\rangle) = (a|010\rangle + b|101\rangle) \otimes |10\rangle$$

$\mathcal{E}_{10}$,

$$R_{10} = \mathcal{E}_{10}^*$$

$$\Rightarrow \mathcal{E}_{10}^*(\mathcal{E}_{10}|\Psi\rangle) = \underbrace{\mathcal{E}_{10}^*\mathcal{E}_{10}}_{= \mathbb{1}}|\Psi\rangle$$

Fact list of Pauli Group

- Pauli Group $\mathcal{P}_n$

1) $\sigma_0, \sigma_1, \sigma_2, \sigma_3 \in \mathcal{L}(\mathbb{C}^2)$. 2×2

$\quad\quad \sigma_1 = \sigma_x,\ \sigma_2 = \sigma_y,\ \sigma_3 = \sigma_z$

properties ① $\sigma^* = \sigma$, $\sigma\sigma^* = \sigma^*\sigma \Rightarrow \sigma^2 = \mathbb{1}$

② $\{\sigma_0, \sigma_1, \sigma_2, \sigma_3\}$ spans $\mathcal{L}(\mathbb{C}^2)$.

③ $\{i^\alpha \sigma_\beta\}_{\substack{\alpha=0,1,2,3 \\ \beta=0,1,2,3}}$ forms a group $\mathcal{P}_1$

2) In general, $\mathcal{L}((\mathbb{C}^2)^{\otimes n})$

$\quad\quad \cong \mathcal{L}(\mathbb{C}^2) \otimes \mathcal{L}(\mathbb{C}^2) \otimes \cdots$

$$\mathcal{P}_n = \{ i^\alpha \sigma_{a_{n-1}} \otimes \sigma_{a_{n-2}} \otimes \cdots \otimes \sigma_{a_0} \}$$

forms a group. $\quad (zv \otimes w = v \otimes zw)$

$$(\sigma_x \otimes \sigma_y) \cdot (\sigma_y \otimes \sigma_z) = \sigma_x\sigma_y \otimes \sigma_y\sigma_z$$

3) ① P_n "spans" $\mathcal{L}((\mathbb{C}^2)^{\otimes n})$

$\forall T \in \mathcal{L}((\mathbb{C}^2)^{\otimes n}), \exists a_j \in \mathbb{C}, h_j \in P_n$

s.t. $T = \sum_{\text{finite}} a_j h_j$

② $\forall g, h \in P_n$, $gh = hg$

or $gh = -hg$.

③ $\omega_p(\sigma_x \otimes \sigma_z \otimes 1) = 2$

$\omega_p(i\sigma_x \otimes \sigma_x \otimes \sigma_y \otimes 1 \otimes \sigma_z) = 4$

$d_p(C_q) = \min\{ \omega_p(g) : g \in P_n$ s.t. $\exists |\Psi_x\rangle, |\Psi_y\rangle$

$\in$ ONB of $\mathcal{H}^{C_q}$,

satisfying $\langle \Psi_x | g \Psi_y \rangle \neq f(g)\,\delta_{xy}$

$\forall f : P_n \to \mathbb{C} \}$.

$\boxed{P_n \curvearrowright \mathcal{H}^n, \ \mathrm{Stab}_{P_n}(\mathcal{H}^{C_q})}$